

Annex B

Cover Page for College to submit with Prelim Examination Paper

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Inequalities	9740/01	$x < -2$ or $x = -\frac{1}{2}$ or $x > 1$	4
2	Curve Sketching	9740/01	-	5
3	Complex Numbers	9740/01	(i) $1-i = \sqrt{2}e^{-\frac{\pi}{4}i}$ (ii) $32-33i$	5
4	Vectors	9740/01	(i) $\mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 8 \end{pmatrix}$ (ii) $\theta = 60^\circ$ (iii) $\frac{21\sqrt{3}}{2}$	7
5	Mathematical Induction/ Recurrence	9740/01	(i) $u_2 = \frac{3}{4}, u_3 = \frac{5}{8}, u_4 = \frac{7}{16}$ (ii) $u_n = \frac{2n-1}{2^n}$	7
6	Differentiation	9740/01	$k=4$	6
7	Method of Difference, AP,GP & Sigma Notation	9740/01	(i) $\frac{1}{2}\left(\frac{1}{r} - \frac{1}{r+2}\right)$ (ii) $\frac{3}{4} - \frac{1}{2}\left(\frac{1}{n+1} + \frac{1}{n+2}\right)$ (iii) $\frac{3}{4}$ (iv) $\frac{89}{840}$	7
8	Differentiation	9740/01	$y - \cos^3 p = -\cot p(x - \sin^3 p)$ $T = (\sin p, 0), N = (0, \cos p)$	8
9	Integration	9740/01	$\sum_{r=1}^n \frac{2}{2n+r}$	8
10	Integration	9740/01	(a) $\frac{2}{3}\pi + \ln\left(\frac{\sqrt{3}-1}{2}\right)$ (b) $\frac{\pi}{3} + \frac{\sqrt{3}}{4}$	9
11	Functions	9740/01	(iii) $a = -\frac{1}{2},$	11

			$h^{-1}: x \mapsto \frac{-1 + \sqrt{4x+9}}{2}, x > -\frac{9}{4}$ $(iv) h^{-1}f: x \mapsto \frac{-1 + \sqrt{4(2^x)+9}}{2}, x \in \mathbb{R}$ $R_{h^{-1}f} = (1, \infty)$	
12	Curve Sketching	9740/01	$a = 1, b = 2, c = 1, d = -5$ (i) one real root (ii) two real roots	11
13	Differential Equations	9740/01	$\frac{dx}{dt} = -kx, 23 \text{ years}$ $x = \frac{1}{8} [4t \tan^{-1} 2t - 2 \ln(1 + 4t^2) + 4]$	12
Pure Maths				
1	Maclaurin's Series	9740/02	-	5
2	Integration	9740/02	(i) $\frac{16}{15}$ (ii) $\frac{44}{15}\pi$	6
3	Complex Numbers	9740/02	(i) $a = 6, b = \sqrt{20}$ (ii) $\min \arg = 0, \max \arg = \tan^{-1}\left(\frac{1}{2}\right)$	8
4	Vectors	9740/02	(ii) $\mathbf{q} = \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$ $\mathbf{p} = \frac{1}{5} \begin{pmatrix} 11 \\ 5 \\ 23 \end{pmatrix}$	9
5	AP, GP & Sigma Notation	9740/02	(a) $3k^2$ (b) (i) 13 (ii) 1666.7	12
Statistics				
6	Sampling, Testing	9740/02	-	4
7	P & C	9740/02	86 400	4
8	Sampling, Testing	9740/02	(a) (ii) 8.46%	8
9	Regression	9740/02	(ii) A (iii) $y = -71.370 + 36.695 \ln x$ (iv) 13.1	9
10	Normal	9740/02	(i) 1000 (ii) 0.705	8

			(iii) $\sigma = 303.97$	
11	Poisson	9740/02	(i) 0.0555 (ii) 0.00158 (iii) 0.408 $k = 28$	14
12	Binomial	9740/02	(i) 0.963 (ii) 0.00280 (iii) 0.999 (iv) 0.882	13